

11. (a) Esters are formed when carboxylic acids are heated with alcohols in the presence of concentrated sulfuric acid. This is a reversible reaction that is carried out by heating the reagents under reflux.

(i) Draw a labelled diagram of the apparatus you would use to carry out a reaction under reflux. [3]

(ii) Explain how the apparatus that you have drawn in part (i) results in the process of reflux. [1]

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(iii) Why is it necessary to use a reflux technique in this type of reaction? [1]

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(iv) State the function of the concentrated sulfuric acid in this reaction. [1]

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- (b) The equation shows the reaction between ethanoic acid and methanol.



When 3.00g of ethanoic acid was heated under reflux with 1.28g of methanol in the presence of concentrated sulfuric acid for 10 minutes it was found that 1.18g of methyl ethanoate was formed.

- (i) Suggest how the methyl ethanoate could be separated from the mixture formed in the reaction vessel. [1]

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- (ii) Calculate the percentage of theoretical yield of methyl ethanoate obtained. Show your working. [3]

Percentage yield = %

- (iii) Suggest **one** change that could be made to this preparation to improve the percentage yield of methyl ethanoate. Explain your suggestion. [2]

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- (c) Two reactions of organic compound **A** are shown below.



Compound **B** is a straight-chain hydrocarbon with the formula C_4H_8 .

- (i) What type of reaction is taking place when compound **A** is heated with concentrated sulfuric acid to form compound **B**? [1]
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- (ii) Draw the displayed formulae of **two** possible isomers of **A**. [2]
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- (iii) What colour **change** is seen when compound **A** reacts with the $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$ solution? [1]
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- (iv) What type of reaction is taking place when compound **A** is heated with $\text{Cr}_2\text{O}_7^{2-}/\text{H}^+$ to form compound **C**? [1]
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- (v) Draw a displayed formula of compound **C**. [1]

